Short-Term Fourier Transform as Preprocessing for Common Spatial Pattern

Ana Beatriz Lana Maciel Moreira Armond
Universidade Federal de Juiz de Fora
Juiz de Fora, Brazil
0009-0006-1156-9261

Gabriel Henrique de Souza
Universidade Federal de Juiz de Fora
Juiz de Fora, Brazil
0000-0002-5931-654X

Davi Esteves dos Santos
Universidade Federal de Juiz de Fora
Juiz de Fora, Brazil
0009-0003-3572-6071

Heder Soares Bernardino
Universidade Federal de Juiz de Fora
Juiz de Fora, Brazil
0000-0003-2012-7802

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***Abstract*—Brain-Computer Interface (BCI) allows direct communication between the human brain and a computer system. Motor Imagery (MI), a prominent BCI paradigm, decodes a mental activity of motor planning in a message to a device. This process can incorporate signal acquisition, preprocessing, temporal and spatial filtering, feature extraction, and classification. Common Spatial Pattern is a well-known method for spatial filtering in BCI. Combining CSP with temporal filtering methods, such as Bandpass and Empirical Mode Decomposition, improves efficiency. This study proposes using CSP in the time-frequency domain with Short-Time Fourier Transform (STFT) as a temporal filter. STFT detects fluctuations in brain activity across time and frequency. It can recognize MI-related elements at various frequencies. We compared STFT-CSP with CSP, Filter Bank CSP, Median-SEE, and Sigmoid-SEE concerning the task of distinguishing between left and right hand movements. The proposed STFT-CSP achieved the best overall result and was more robust than the other approaches. Moreover, by analysing the performance profiles, we can conclude that in the worst-case scenario, the accuracy was improved in more than 30%. Therefore, the method increased result stability across the subjects. The proposed STFT-CSP proved to be more consistent than other approaches and more suitable for real-world applications.**

***Index Terms*—Electroencephalogram, Motor Imagery, Filter Bank, Temporal filtering.**

I. Introduction

Brain-computer interface (BCI) has been advancing significantly on new approaches for decoding brain activity for real-world applications [1]. Motor Imagery (MI) stands out among many paradigms [2] allowing the classification of imagined movements. MI is used in applications such as control of prostheses [3] and post-stroke motor rehabilitation [4]. Non-invasive acquisition methods like Electroencephalography (EEG) have become crucial in the MI paradigm [5].

The BCI pipeline is composed of four steps [6]: (i) Preprocessing; (ii) Temporal and spatial filtering; (iii) Feature extraction and selection; and (iv) Classification. Depending on the paradigm, different methods can be used for each step in the pipeline. MI is one of the most used paradigms for BCI [7] and is no exception to the aforementioned pipeline. Two common methods in MI-BCI pipelines are Common Spatial Pattern (CSP) [8] and Filter Bank [9].

The temporal filtering step uses temporal variation to extract relevant information from brain signals, such as event-related potentials (ERPs) or oscillatory activities at various frequencies. Some well-known temporal filters are Empirical Mode Decomposition (EMD), Band-pass filtering (BP), and Short-Time Fourier Transform (STFT) [10]. EMD is an adaptive decomposition approach that separates the signal into intrinsic modes components, and BP filters the EEG signal in a set of frequency bands. EMD and BP are well-known approaches in the literature, although they have certain limitations. Temporal filtering may be limited when dealing with non-stationary signals, such as those with abrupt changes. Moreover, EMD [11] does not create features based on the frequencies, losing some significant components.

STFT analyzes changes in brain activity over time in a set of frequencies. It can find features related to motor imagery at various frequency ranges. Therefore, STFT can improve the BCI results when using other techniques for spatial filters, such as CSP. In this work, we propose STFT-CSP and perform experiments to validate that hypothesis.

In the experiments, we could see that STFT-CSP performed with high stability, returning good accuracies in varied datasets and providing the best-worst case in two of the experiments, when evaluated against Filter Bank CSP (FBCSP) [12], CSP, MedianSEE, and SigmoidSEE [13].

The remainder of this paper has the following sections: In Section II, we present related works in the literature of STFT in BCI pipelines and CSP; Section III presents an explanation of the methods used; Section IV describes the proposed STFT-CSP; Section V shows the datasets used in this work; Section VI discusses the performed experiments and shows the results and their analysis; and Section VII concludes this paper and proposes future works.

II. Related Works

Temporal Filtering techniques can improve existing methods by enhancing the signal-to-noise ratio (SNR) and feature relevance. There are different ways to use STFT in BCI pipelines, such as temporal filtering. In this scenario, STFT yielded relevant results to differentiate subject resting-state and hand-imagery states [14] and to extract features in the time-frequency domain [15]. Moreover, STFT has been used to extract features of Magnetoencephalography (MEG) [16], which pointed out that STFT has relevant results with different signal acquisition methods. STFT with CSP was also used for event-related desynchronization (ERD) [17].

Spatial filtering is a critical technique in BCI, used to increase or decrease signals based on their spatial locations and reduce the interference between electrodes. CSP [8] is a well-known spatial filter for BCI [12], [18], [19]. This method uses spatial projections that maximize variance for one class while minimizing variance for another.

Usually, temporal filtering is used before CSP, creating many combinations of methods. For instance, when CSP is after a bandpass or a Filter Bank (FB), a set of bandpass filters. FBCSP [9], [12], [20] is a well-known BCI pipeline that divides the EEG signal into many bands and applies CSP in parallel for each band. STFT was also used for the MI paradigm in a novel approach using multidimensional Empirical Mode Decomposition (MEMD) [21]. Moreover, Convolutional Neural Networks (CNN) with STFT improves their results [22].

This work aims to improve the standard CSP pipeline by employing STFT, which increases feature relevance in the time-frequency domain. STFT-CSP combines STFT's temporal and frequency resolutions with CSP's spatial discrimination capabilities. The properties of the proposed method produced good and more stable outcomes, and stability distinguishes it from the existing literature.

III. Methods

This section describes and explains the methods employed for our proposal and experiments. Here, there are the descriptions of two temporal filter methods: bandpass and Filter Bank; a spatial filter: CSP; a Feature Selection: Mutual Information-based Best Individual Feature (MIBIF); and Naive-Bayes Parzen Window (NBPW) classifier.

A. Band-pass Filter

Bandpass filtering [20] in BCI is a technique that extracts the frequencies of the brain's signals correlated with the brain's functions. When the bandpass filter is applied, the new signal has a better SNR and is more correlated to the expected output of the system.

B. Filter Bank

Filter Bank (FB) [9] is a set of bandpass filters applied to different frequencies. FB divides the EEG signal into frequency bands to select specific information associated with different brain activities. FB presents better results than the bandpass for most cases since it decomposes the original signal in many sub-bands. Therefore, it increases the SNR and allows the use of the bands separately.

C. Short-time Fourier Transform

The Short-time Fourier Transform (STFT) is a technique to analyze the signal's frequencies through time. For that, it applies a Fourier transform with a sliding window. The discrete approach of the method is related to the discrete-time Fourier-Transform (DFT) [10], given by:

$$X(\omega) = \sum_{n=\infty}^{\infty} x(n) e^{-j\omega n} \tag{1}$$

where $X(\omega)$ represents the Fourier Transform of the signal at frequency ω. $x(n)$ is the input sequence in the discrete-time domain (signal input), and n is the discrete-time variable.

The output of STFT is a set of DFTs corresponding to different cropped windows of x. The expression to the discrete-time STFT at time t is

$$X(t, \omega) = \sum_{n=\infty}^{\infty} x(n) \phi(t-n) e^{-j\omega n} \tag{2}$$

where $\phi(n)$ is the analysis filter, and the time section for each time t is given by $x(n)$ times the shifted sequence $\phi(t-n)$.

However, while dealing with a digital signal inherently characterized by discrete sampling, it is necessary to use the discrete STFT by applying the DFT over a finite-sequence of time:

$$X(n, \omega) = \sum_{m=\infty}^{\infty} x(m) \phi(n-m) e^{-j2\pi km/N} R_N(k) \tag{3}$$

where $X(n, \phi)$ is the representation of the STFT in time n and frequency ω, $x(m)$ is the sample of the signal in discrete time m, and $\phi(n-m)$ is the window function applied to the signal. Moreover, $R_N(k)$ is a normalization function that depends on the size of the DFT N, and $e^{-j2\pi km/N}$ represents the frequency contribution to the DFT for a given k, which is the index that cycles through the discrete frequencies corresponding to the DFT.

Thus, STFT applies DFT to finite time sequences, enabling exact analysis of signal frequencies over discrete intervals. This technology is vital for extracting usable information from digital signals, which improves characteristics in both time and frequency domains.

D. Common Spatial Pattern

CSP aims to find a linear transformation W that maximizes the distance between two classes while decreasing the variability intra-group [23].

The application of W on the data is defined by:

$$Z_i = W^T \times E_i \tag{4}$$

where $E_i \in \mathbb{R}^{S \times Ts}$ represents the i-th signal, S is the number of electrodes, Ts is the size of the signal collected per electrode in each trial, $W \in \mathbb{R}^{S \times S}$ is the transformation matrix, and $Z_i \in \mathbb{R}^{S \times Ts}$ is a matrix obtained by the application of the CSP linear transformation.

CSP aims to maximize the variance difference between correlation matrices between electrodes in each class [8].

The correlation matrix for electrodes of a class c is used to maximize the difference. That matrix is defined as

$$\Sigma^{(c)} = \frac{1}{N_c} \sum_{i=1}^{N_c} E_i^{(c)} \times (E_i^{(c)})^T, \tag{5}$$

where $\Sigma^{(c)} \in \mathbb{R}^{S \times S}$ is the correlation matrix between the electrodes for class c, N_c is the number of training samples of the class c, and $E_i^{(c)}$ is the i-th training sample of the class c.

Then, the correlation matrix is used to calculate W presented in Equation 4 by the simultaneous diagonalization of the two correlation matrices as

$$W^T \Sigma^{(1)} W = \Lambda^{(1)}, \quad \text{and} \quad W^T \Sigma^{(2)} W = \Lambda^{(2)}, \tag{6}$$

Finally, the LogPower function is applied to the transformed signal Z to extract the features from the time-space as

$$f_i = \log \left(\frac{\text{diagonal}(Z_i \times Z_i^T)}{\text{trace}(Z_i \times Z_i^T)} \right), \tag{7}$$

where f_i represents the feature vector of Z_i, and diagonal() and trace() return, respectively, the main diagonal and the trace of a matrix. Moreover, only the first and last $\frac{m}{2}$ features are used since it has the most difference between the classes.

E. Mutual information-based best individual feature

Mutual Information-based Best Individual Feature (MIBIF) is a feature selection method that selects the best characteristics by considering mutual information across features and classes. [19], [24], [25]. This method sorts the characteristics based on mutual information in decreasing order as

$$I(X;Y) = H(Y) - H(Y|X_j) \tag{8}$$

Where $X \in \mathbb{R}^{N \times F}$ is the training set with N trials and F dimensions, $H(Y) = -\sum_{w=1}^{L} P(w) \cdot \log_2 P(w)$ is the entropy of Y. The conditional entropy $H(Y|X_T^j)$ of Y given X_T^j is calculated as

$$H(Y|X) = -\sum_{w=1}^{L} \sum_{i=1}^{N_w} P(w|X)_{i,j} \cdot \log_2[P(w|X)_{i,j}] \tag{9}$$

where, $P(w|X_{i,j}) = \frac{P(w) \cdot P(X_{i,j}|w)}{P(X_{i,j})}$ is the conditional probability of w given $X_{i,j}$ defined using Bayes' theorem. N_w is the number of samples of class w, and $X_{(w)i,j}$ is the value of the j-th feature of the i-th sample of class w.

F. Naive Bayes Parzen Window

Naive Bayes Parzen Window (NBPW) classifier [19] is a method that calculates the probability of class w, $P(w|X_i)$. The probability is calculated using Bayes' theorem and the assumption that the samples are independent as

$$P(w|X_i) = \frac{P(w) \prod_{j=1}^{J} P(X_{i,j}|w)}{\prod_{j=1}^{J} P(X_{i,j})} \tag{10}$$

where $P(X_{i,j}) = \sum_{w=1}^{L} P(w) \cdot P(X_{i,j}|w)$. Moreover, the probability $P(X_{i,j}|w)$ is calculated by Parzen Window [26] as

$$P(X_{i,j}|w) = \frac{1}{N_w} \sum_{k=1}^{N_w} \phi(X_{i,j} - X_{k,j}, h) \tag{11}$$

where $\phi(y,h)$ is a smoothing kernel function, which can be calculated as

$$\phi(y,h) = \frac{1}{\sqrt{2\pi} \cdot e^{-\frac{y^2}{2h^2}}} \tag{12}$$

where σ is the standard deviation and $h = \frac{4}{3N_w} \times 0.2\sigma$, as proposed by Bowman and Azzalini [27].

G. Filter Bank Common Spatial Pattern

Filter Bank Common Spatial Pattern (FBCSP) [12] algorithm is a classical pipeline for BCI applications. The main stages of FBCSP are: (i) A FB with nine bands: 4–8, 8–12, ..., 36–40 Hz; (ii) CSP as spatial filter; (iii) MIBIF as feature selection; and (iv) A classification using NBPW. Moreover, FBCSP steps can be seen in a schematic way in Figure 1.

IV. Proposed Short-Term Fourier Transform with Common Spatial Pattern

This work proposes the discrete STFT as a temporal filter and uses the CSP in the time-frequency domain. CSP is used mainly in the time domain, even with FB, the method still loses some frequency features. Similarly to FB, STFP separates the signal into many sub-bands, though with a more precise frequency sampling. Then, we performed experiments for the STFT-CSP model in MI applications and found promising results as presented in Section VI. After CSP, we used MIBIF as a feature selector and NBPW as a classifier. However, any classifier is feasible in this step, including deep learning techniques such as EEGNet [28]. Figure 2 shows the schematic approach of STFT-CSP.

V. Datasets

This section presents the datasets for MI that we used in the experiments. We chose them because of their relevance and widespread use in BCI literature. Moreover, all datasets presented here have the Motor Imagery paradigm and were recorded with EEG. Figure 3 shows the 10-10 system for electrode position that was used for all datasets.

A. Clinical BCI Challenge Dataset

The Clinical BCI Challenge (CBCIC) Dataset [30] has the data of 9 hemiparetic subjects. The subjects had limited mobility caused by a stroke and treated that limitation using a BCI system. Each subject completed 80 trials during training and 40 trials during testing. Trials take 8 seconds, with a random 2 to 3-second blank screen before the new trial begins. The subject hears a "get ready" message for the first three seconds, followed by a warning sound after two seconds. A cue is given starting at 3 seconds and lasting 8 seconds. The data was recorded using 12 electrodes (F3, FC3, C3, CP3, P3, FCz, CPz, P4, FC4, C4, CP4, and P4) and a sampling rate

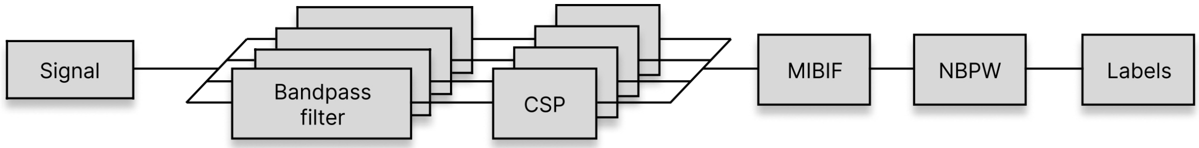

Fig. 1. FBCSP architecture [12].

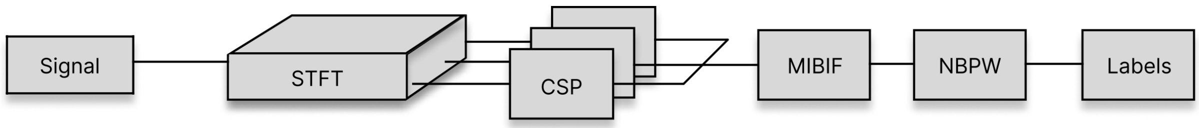

Fig. 2. Proposed STFT-CSP Approach.

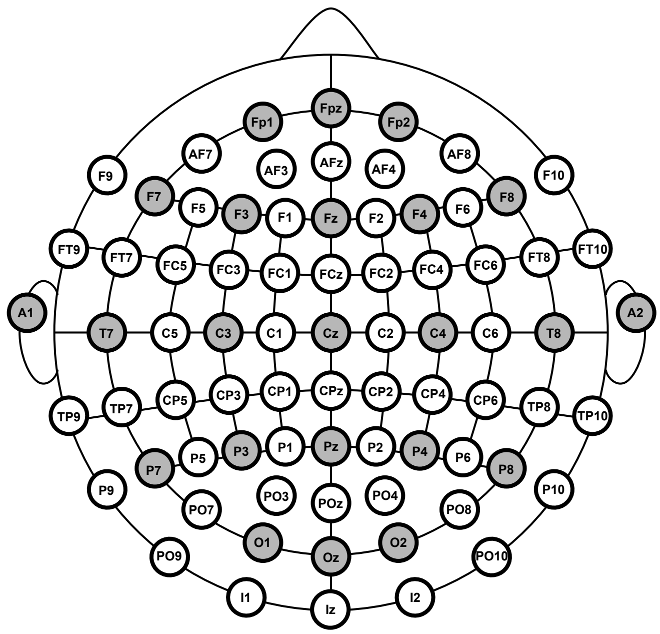

Fig. 3. 10-10 system for electrode. Modified from [29].

of 512 hertz. Moreover, the signal had a bandpass filter of 0.5-100 Hz implemented in hardware and a notch filter with a frequency range of 50 Hz.

B. BCI competition IV 2a

The dataset 2a from BCI competition IV (BCICIV2a) [31] is a well-known dataset in BCI. It was established in 2008 as part of the fourth BCI competition and have motor imagery tasks. The data were recording from 9 subjects at a sample rate of 250 Hz. The signal was collected using 22 electrodes (Fz, FC3, FC1, FCz, FC2, FC4, C5, C3, C1, Cz, C2, C4, C6, CP3, CP1, CPz, CP2, CP4, P1, Pz, P2, POz) in two days with 288 trials each. The trials last 6 seconds and are followed by a short break. The trial begins with a warning sound and displays a fixation cross on the screen for 2 seconds before a 1.25-second cue. There are four possible classes: left-hand, right-hand, tongue, and both-feet. A bandpass filter with a frequency range of 0.5-100 Hz and a notch filter was using during the record.

C. BCI competition IV 2b

The dataset 2b (BCICIV2b) [31] is another presented in BCI Competition IV, such as the dataset in Section V-B. It is also an MI dataset but with just two classes: left-hand and right-hand. The data were recorded from 9 subjects using three electrodes (C3, Cz, C4) with a sampling rate of 250 Hz. A bandpass filter of 0.5-100 Hz was applied in hardware, and also a 50 Hz notch filter. The trials last 8 seconds, followed by a brief break. Five sessions were recorded per subject on different days when the first two sections were without feedback, and the others were executed with feedback. The trial protocol for the first two sessions includes: (i) a fixation cross on the screen for 3 seconds; (ii) a warning sound at 2 seconds; and (iii) a 1.25-second cue. The trial methodology for the last three sessions includes: (i) a gray smiley feedback face for the first 3.5 seconds; (ii) a warning sound at 2 seconds; and (iii) a green or red smiley feedback face for the next 5.5 seconds. The first two sessions consisted of 120 trials, whereas the following three sessions consisted of 160 trials divided equally among them.

VI. Computational Experiments

This session presents the settings we used in the experiments and its results. The following experiments were performed on the datasets presented in Section V. The code used in this work is available at https://github.com/LanaArmond/STFT-CSP. The BCICIV2a has four possible classes for the trials, but we used only two classes: left-hand and right-hand. We reduced it since most applications for MI uses only left-hand and right-hand. We employed a 5-fold stratified cross-validation strategy, training models 0.5s after the cue. The EEG signal was downsampled to 128 Hz as in [28] and filtered with a bandpass filter from 4 to 40 Hz. For the CSP and FBCSP methods, we used $m = 2$ pairs of features and eighth features selected on MIBIF as in [20], while the STFT with CSP (STFT-CSP) utilized segments of length 128 and a Hann window. MedianSEE and SigmoidSEE used the single electrode energy

(SEE) method to classify clinical data, with parameters used as specified by [13].

To obtain a more insightful comparison of the models, we used performance profiles. It is necessary to define a set of problems P, which will be used to evaluate performances as in [32], and a set of methods S, which will be the five already presented, for the performance profiles. The selected 27 problems per method are the average accuracy for the five folds of the nine subjects from the three datasets. Performance profiles (PPs) [33] is a visual tool for the analysis of the relative performance of a set of methods (S) when solving problems from a set P. In this study, S is composed of the five methods analyzed here and P is formed of the 27 problems (3 datasets and 9 subjects). The performance metric used is the average accuracy obtained in the five folds. The performance profiles highlight three key insights [32]: (i) the approach with the largest $\rho(1)$ is the one that achieved the best results; (ii) the approach with the smaller tau that reaches $\rho(\tau) = 1$ is the most robust; and (iii) the approach with the largest area under the curve is the best overall approach.

The tables in this section present the accuracy of the methods for each subject, with the last row showing the average for all subjects. Moreover, we used the Kruskal-Wallis test as a group test with Dunn's test with Bonferroni correction as a post-hoc test. The values marked with the symbol (*) are statistically relevant since they yielded a p-value ≤ 0.05 in correlation with STFT-CSP. The results statistically different from those obtained by STFT-CSP (p-value ≤ 0.05) are marked with a star (*).

A. Real-work application

We used the CBCIC dataset to evaluate the methods in a real-work application for BCI since its data is for post-stroke motor rehabilitation. Table I presents the results of this dataset. The best result was found by CSP, followed by the STFT-CSP, with no statistical difference between them. Moreover, STFT-CSP had the best-worst case, obtaining 0.7 from subject 8. In the last row, we can see that FBCSP, MedianSEE, and SigmoidSEE had statistically different outcomes than STFT-CSP. Moreover, the STFT-CSP had the best result for 5 of 9 subjects.

TABLE I
ACCURACY RESULTS FOR THE CBCIC DATASET. (*) RESULTS STATISTICALLY DIFFERENT OF STFT-CSP.

Subject	CSP	FBCSP	MedianSEE	SigmoidSEE	STFT-CSP
1	0.7417	0.6917	0.7417	0.8000	**0.7667**
2	0.8250	0.5750	0.6417	0.6500	**0.8417**
3	0.7417	0.5417	0.7167	0.5750	**0.7667**
4	**0.8167**	0.5083	0.5667	0.6417	0.7583
5	**0.8250**	0.5333	0.7500	0.7083	0.7250
6	**0.7667**	0.5833	0.7000	0.7417	0.7417
7	0.6917	0.5333	0.6667	0.6500	**0.7750**
8	**0.7083**	0.5333	0.7000	0.6833	0.7000
9	**0.7417**	0.4583	0.5917	0.4667	**0.7417**
all	**0.7620**	0.5509*	0.6750*	0.6574*	0.7574

B. Many electrodes case

The dataset BCICIV2a was used to evaluate the approach for many electrodes. That is an important experiment, since CSP performs better with higher electrode density. Table II shows the results and pointed out that FBCSP performed better than other methods. FBCSP had an average accuracy of 0.8318, and it had the best result for 8 of 9 subjects. Moreover, it was statistically different than the others. STFT-CSP was the second best model, and it was statistically different of the other models except CSP.

TABLE II
ACCURACY RESULTS FOR THE BCICIV2A DATASET. (*) RESULTS STATISTICALLY DIFFERENT OF STFT-CSP.

Subject	CSP	FBCSP	MedianSEE	SigmoidSEE	STFT-CSP
1	0.7743	**0.9306**	0.5694	0.5729	0.8438
2	0.5556	**0.5799**	0.5278	0.5208	0.5000
3	0.9375	**0.9514**	0.5764	0.5799	0.8750
4	0.6597	**0.7083**	0.5104	0.4653	0.6562
5	0.5868	**0.8889**	0.5243	0.5382	0.7708
6	**0.6146**	0.5833	0.5000	0.5312	0.6042
7	0.7465	**0.9583**	0.5243	0.4653	0.8264
8	0.9514	**0.9549**	0.6389	0.6389	0.9167
9	0.9306	**0.9306**	0.7049	0.7188	0.8958
all	0.7508	**0.8318***	0.5640*	0.5590*	0.7654

C. Few electrodes and many sessions

That experiment evaluates the approach in an advantage case. The dataset BCICIV2b has just three electrodes, and the data were recorded in five days. This way, CSP has few electrodes to use and has a cross-session problem since the brain patterns can change between sessions. Table III presents the results for this case and points to STFT-CSP as the best model. STFT-CSP had an average accuracy of 0.7402 and was the better method for 4 out of 9 subjects. Moreover, it was statistically different for CSP, Median-SEE, and Sigmoid-SEE.

TABLE III
ACCURACY RESULTS FOR THE BCICIV2B DATASET. (*) RESULTS STATISTICALLY DIFFERENT OF STFT-CSP.

Subject	CSP	FBCSP	MedianSEE	SigmoidSEE	STFT-CSP
1	0.6556	0.675	0.5875	0.5875	**0.7250**
2	**0.5735**	0.5559	0.5691	**0.5735**	0.5632
3	0.4944	0.5347	0.5361	0.5375	**0.5458**
4	0.9203	**0.9297**	0.7905	0.7919	0.9284
5	0.6757	0.8527	0.5500	0.5473	**0.8554**
6	0.6681	0.7667	0.5986	0.5931	**0.7806**
7	0.6667	**0.7292**	0.5861	0.5847	0.7083
8	0.775	**0.7711**	0.7487	0.7513	0.7553
9	0.7431	**0.7819**	0.6264	0.6375	0.7806
all	0.6877*	0.7353	0.6229*	0.6241*	**0.7402**

D. Discussion

STFT-CSP presented the best result for a few electrodes and many session cases. It pointed out that STFT overcame the difficulties caused by the shortcomings of CSP. Moreover, the STFT-CSP was the only method that performed well in

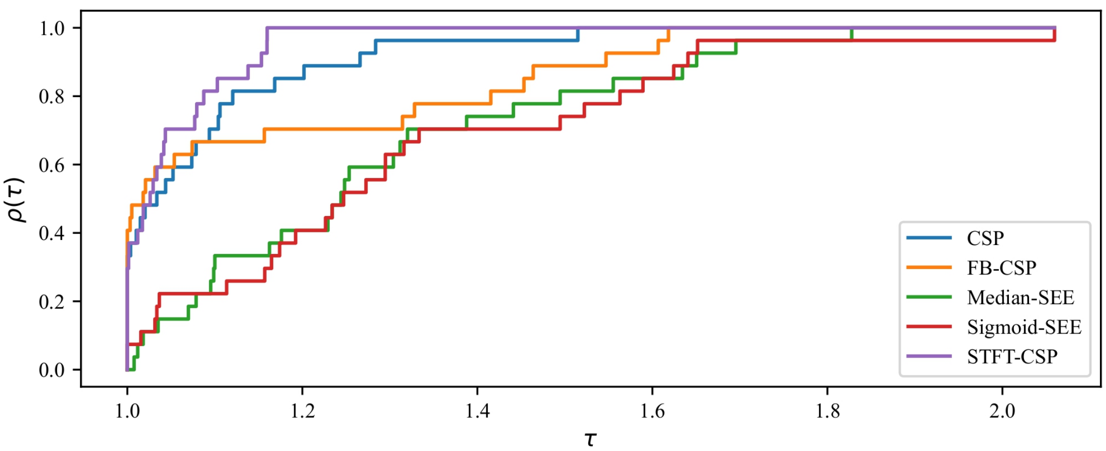

Fig. 4. Performance Profile of the evaluated Methods for all the datasets presented.

all datasets, with average accuracy above 0.7 in all experiments. To present the results more effectively, we used the Performance Profile presented in Figure 4. Considering each subject of all datasets as a problem, FBCSP had the best result for most cases. However, it was the third-best overall method since it had a relevant decrease when it was not the best model. STFT-CSP was the best overall method since it had the largest area in the Performance Profile. Furthermore, in the best-worst case analysis, which is the comparison of the values of τ when the models reach $\rho(\tau) = 1$, STFT-CSP was more than 30% better than the others. In short, the proposed STFT-CSP was the third method with the largerst accuracy per subject, the best overall result, and the best-worst case. Then, STFT-CSP proved the most robust method evaluated and error-resistant. It is relevant in many BCI applications where the number of subjects should be as large as possible. For instance, in post-stroke motor rehabilitation, FBCSP should be the best model for some patients, but STFT-CSP would have sufficient accuracy for more patients.

VII. Conclusion

Non-invasive EEG-based Brain-Computer Interface (BCI) in the motor imagery (MI) paradigm has many applications. The ability to distinguish between specific movements, such as those of the right and left hands, is critical for MI-BCI. To distinguish it, there are many processes in BCI pipelines since the signal has a poor signal-to-noise ratio (SNR). Common Spatial Pattern (CSP) is a well-known method for spatial filtering in BCI pipelines, and it is combined with many temporal filters, such as bandpass and Filter Bank. In this work, we proposed the STFT-CSP approach, which uses the Short-time Fourier transform as a temporal filter before CSP for feature extraction. STFT changes the domain of the problem to time-frequency, giving a more thorough foundation for subsequent analysis.

We evaluated the proposed STFT-CSP in three cases: real-world application, many electrodes, and few electrodes with many sessions. STFT-CSP proved to be the best overall method. Additionally, it had the best worst-case scenario by more than 30% and it was the most robust method. Our research reveals that integrating the STFT as a temporal filtering step before CSP presents a promising strategy for enhancing the accuracy of BCI methods, particularly within MI, based on tests and analysis.

For future works, additional experiments using other spatial filters and classifiers with STFT should be performed. Furthermore, experiments using feedback or other paradigms can point out new ways to use STFT as a temporal filter in BCI problems.

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